

telephones are being progressively installed in the rear of the tanks. The associated commander can directly communicate with the tank's commander through these phones.

With the development of communication technology, tanks are better able to share information. In modern tanks, the commander has a touch-screen interactive display that shows the map of the terrain he is operating in. This is frequently updated by satellites, which helps the tank commander to view the graphical disposition of his own and known enemy troops in the battlefield. He also gets on-field intelligence through unmanned aerial vehicles that transmit live video or image feeds to the tanks. The tank forces—the main strike forces—are slowly switching towards network-enabled capabilities. Through this they share information, designate targets and neutralise them very effectively in close cohesion.

Communication systems for SPG

When front-line troops encounter strong

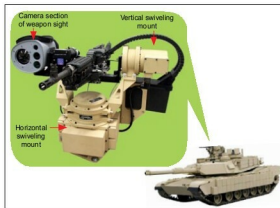


Fig. 18: M1 Tank with Crow's RWS using A M-240 machine gun (Photograph credit: US Army, through Wikipedia. Open domain image modified for use)

resistance from the enemy, they call for an artillery strike. A trooper, known as the forward observer (FO), is present among the front line troops. The FO generates that request through a radio to the fire direction centre (FDC), the overall command centre for SPGs throughout the battlefield. Six to eight guns are grouped to form what is known as a battery, which is commanded by a battery command centre

(BCC). Firing commands from the FDC are issued to the appropriate BCC. The FO, FDC and the SPGs are all situated far away from each other.

In a typical mission, the FO generates the request for an artillery strike to the FDC. He intimates his precise location on the map and the target's location with respect to his location.

The FDC determines which battery would be able to engage the target most ef-

fectively and communicates the same to the BCC. It also determines which shell type has to be used, the required numbers, the elevation and orientation of the gun, etc. In the BCC, guns are directed accordingly to load, align and fire. As the firing is initiated, the FO actively reports to the FDC information, like impact of the shell, side to which the target shell landed and its distance. From these reports, the FDC deduces

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